

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 8

Female Reproductive System

Introduction (Slide 1 of 3)

Sexual reproduction

- This is the union of the nuclei of ovum and sperm cell.
- Ovum and sperm cells are called gametes.
- Each gamete contains half the number of chromosomes of a normal body cell.

Introduction (Slide 2 of 3)

Gametes (*ovum and sperm cell*)

- Gametes are produced in the gonads.
- Female gonads are the ovaries; male gonads are the testes.

Introduction (Slide 3 of 3)

Fertilization

- An ovum leaves the ovaries and travels through the fallopian tube.
- Fertilization takes place if sperm cells are present and unite with the ovum.
- The fertilized egg implants then divides to form a ball of cells, called a zygote (and later an embryo and then a fetus).

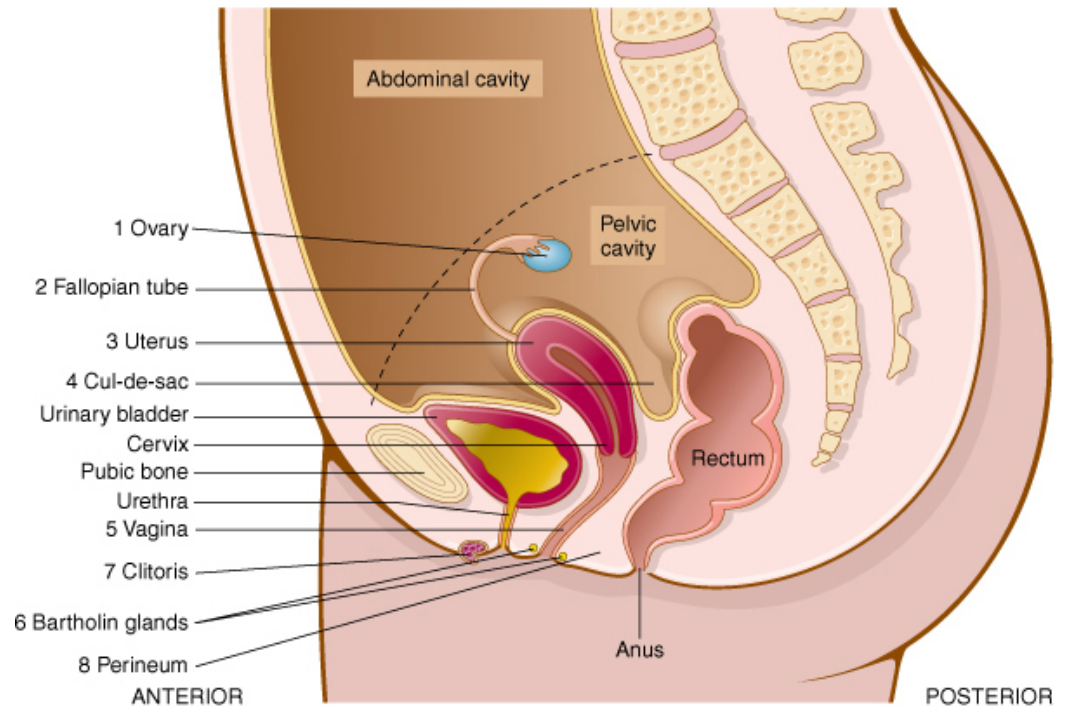
Organs of the Female Reproductive System (slide 1 of 6)

Organs of the female reproductive system (lateral view):

- Ovaries
- Fallopian tubes
- Uterus
- Cul-de-sac
- Vagina
- Bartholin glands
- Clitoris
- Perineum

Organs of the Female Reproductive System (slide 2 of 6)

- Organs of the female reproductive system (lateral view)



QUICK QUIZ (Slide 1 of 5)

1. The fertilized egg implants, then divides to form a ball of cells, called a(n):

- A. embryo
- B. fetus
- C. follicle
- D. zygote

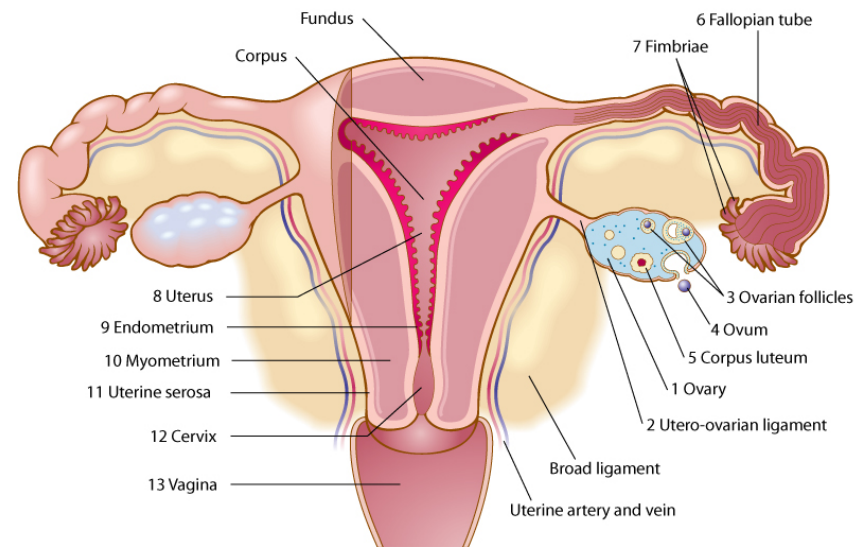
Organs of the Female Reproductive System (slide 3 of 6)

- **Organs of the female reproductive system (anterior view):**

- Ovary
- Utero-ovarian ligament
- Ovarian follicles
- Ovum
- Corpus luteum
- Fallopian tube
- Fimbriae
- Uterus
- Endometrium
- Myometrium
- Uterine serosa
- Cervix
- Vagina

Organs of the Female Reproduction System (slide 4 of 6)

Organs of the female reproductive
system (anterior view)



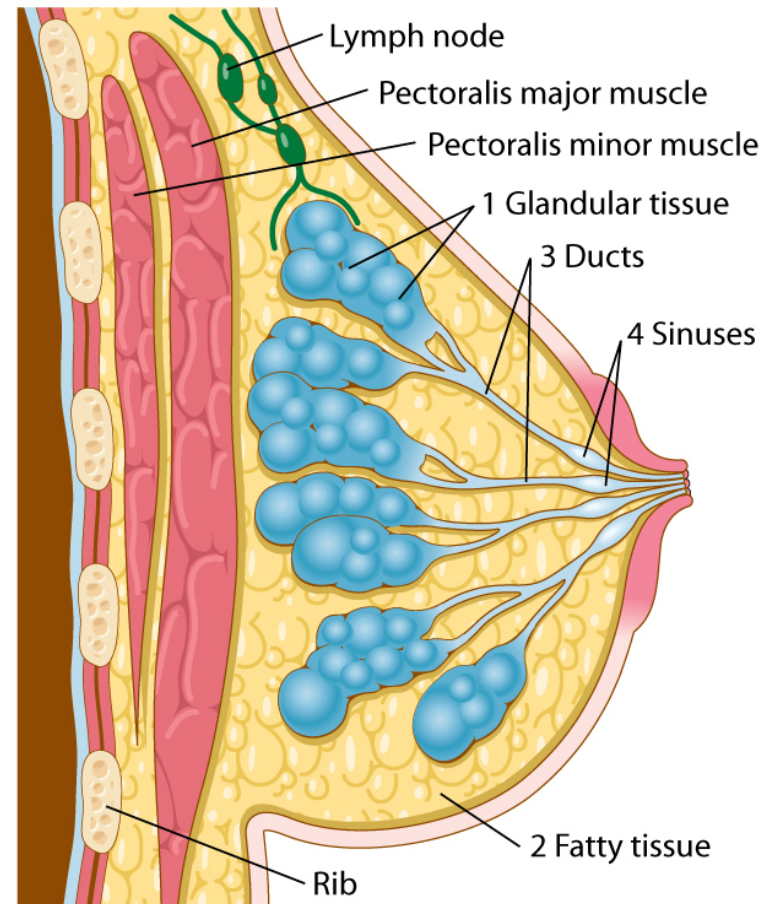
Organs of the Female Reproductive System (slide 5 of 6)

Anatomic structures of the breast (sagittal view):

- Glandular tissue
- Fatty tissue
- Ducts
- Sinuses

Organs of the Female Reproductive System (slide 6 of 6)

The breast (sagittal view)



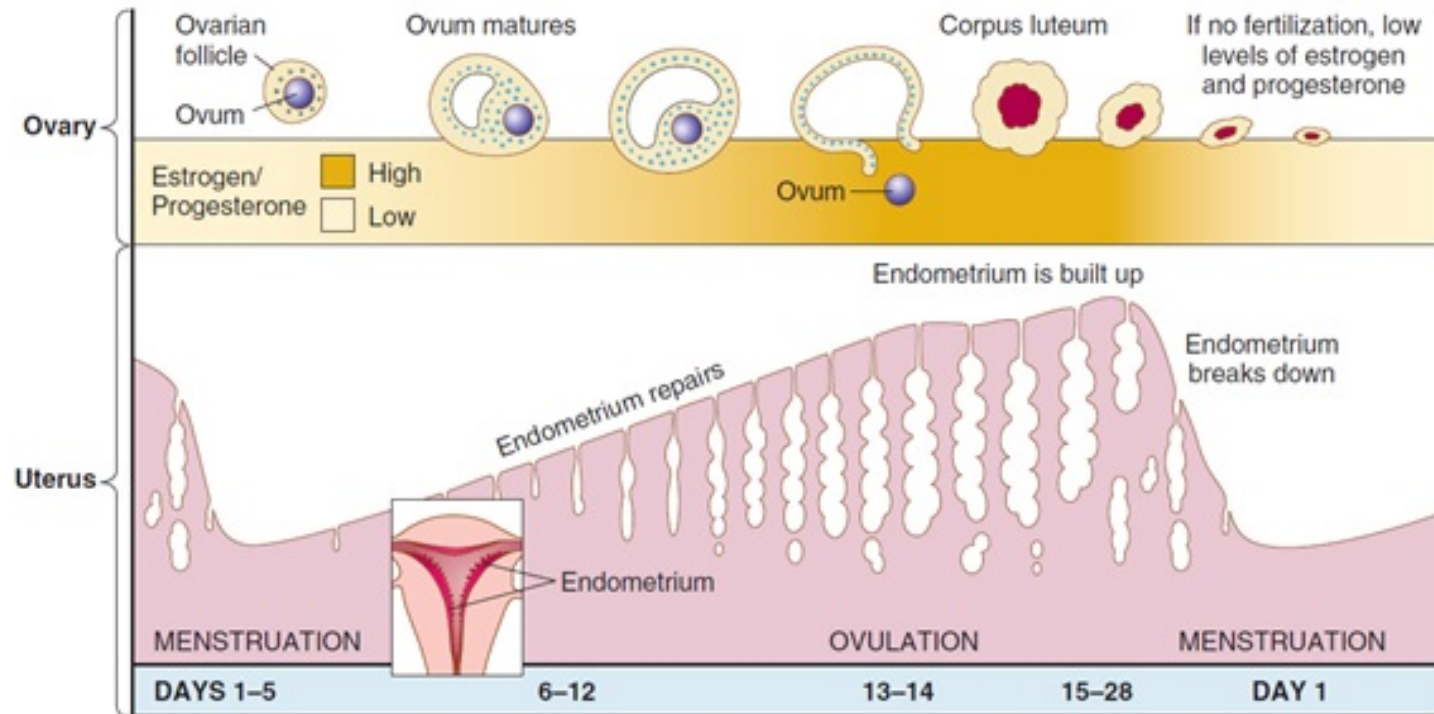
Menstruation (slide 1 of 2)

Menstrual cycle

- **Day 1-5** Blood cells, endometrial cells, and glandular secretions discharge from the body.
- **Day 6-12** Estrogen aids repair of endometrium.
The ovum grows in the follicle.
- **Day 13-14** The egg leaves the ovary and passes through the fallopian tube.
- **Day 15-28** The corpus luteum secretes progesterone;
the uterus lining builds up.

Menstruation (slide 2 of 2)

- Menstrual cycle**



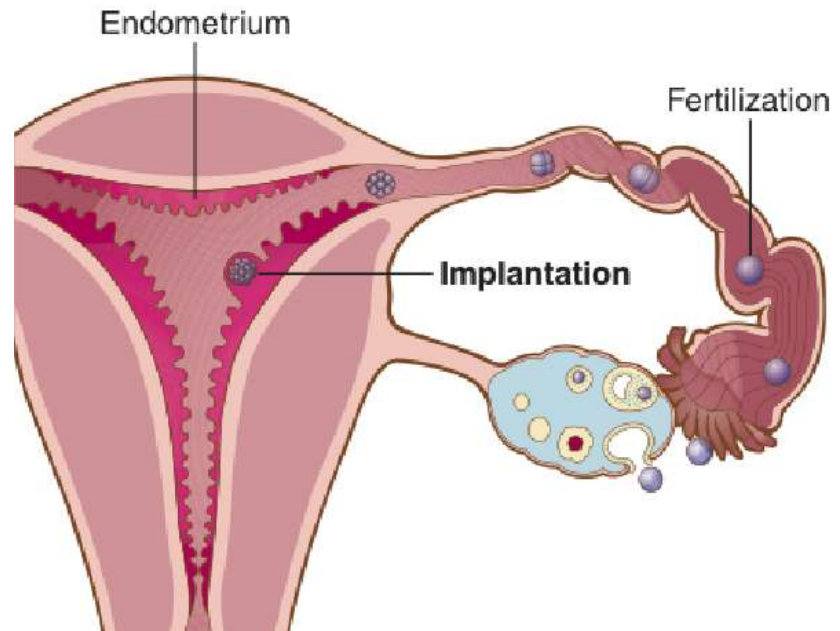
Pregnancy (slide 1 of 6)

Pregnancy

- If fertilization occurs, the egg implants in the uterine endometrium.
- The placenta forms, attaches to the uterine wall.
- Amnion holds the fetus in an amniotic cavity.
- Nutrients, oxygen, and wastes exchange between the fetus and mother.
- The placenta produces human chorionic gonadotropin (hCG).

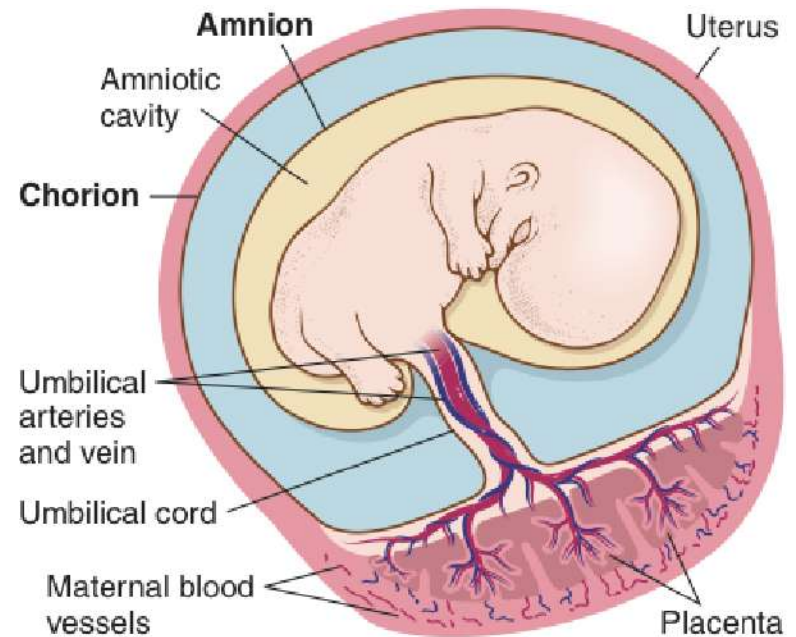
Pregnancy (slide 2 of 6)

- Implantation in the uterus



Pregnancy (slide 3 of 6)

- The placenta, chorion, and amnion membranes



QUICK QUIZ (Slide 2 of 5)

2. The corpus luteum secretes this monthly to build up the lining of the uterus.

- A. Adrenaline
- B. Estrogen
- C. Progesterone
- D. hCG

QUICK QUIZ (Slide 3 of 5)

3. What does the placenta secrete during pregnancy?

- A. Adrenaline
- B. hCG
- C. Progesterone
- D. Estrogen

Pregnancy (slide 4 of 6)

Pregnancy

- Progesterone maintains placenta development.
- The uterus expands as the fetus grows.
- Normal delivery fetal position is cephalic presentation (head first).

Pregnancy (slide 5 of 6)

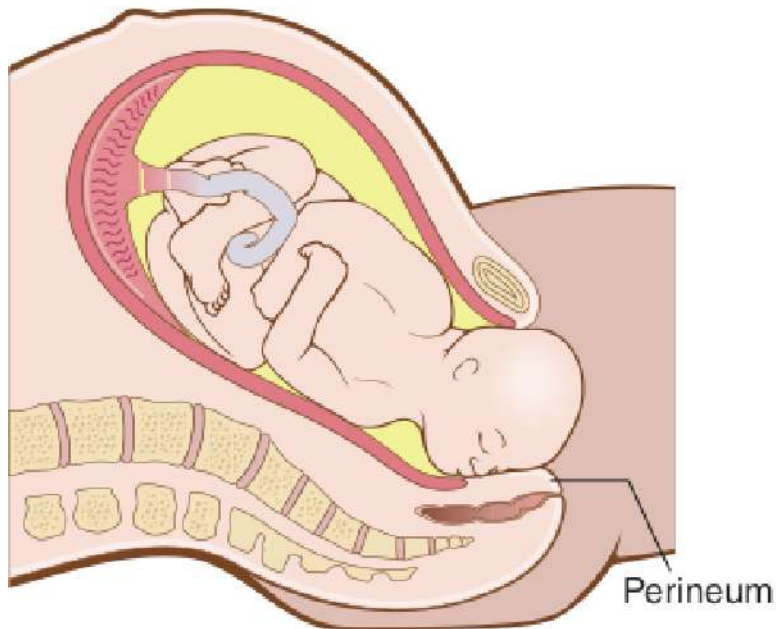
Pregnancy

➤ **Three phases of labor**

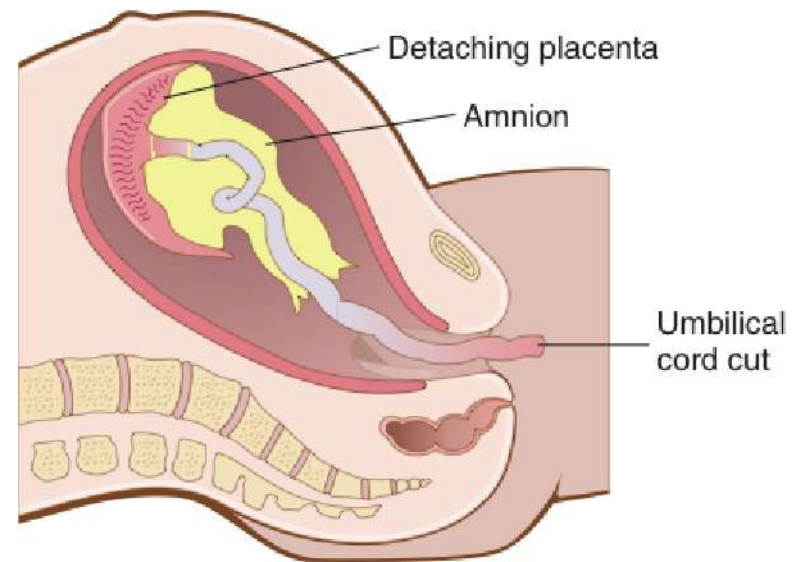
1. Dilation and thinning of the cervix
2. Birth of the infant
3. Delivery of the placenta

Pregnancy (slide 6 of 6)

A, Cephalic presentation.



B, Delivery of the placenta.



Hormonal Interactions

(Slide 1 of 2)

- The pituitary secretes:
 - Follicle-stimulating hormone (FSH)
 - Luteinizing hormone (LH)
- The pituitary stops producing FSH and LH
 - During pregnancy
 - With oral contraceptives

Hormonal Interactions

(Slide 2 of 2)

- Menopause
 - Gradual ending of the menstrual cycle
 - Normally begins between ages 45 and 55
 - Falling estrogen levels
 - Option for estrogen replacement therapy (careful consideration of risks required)

Vocabulary (Slide 1 of 34)

Term
adnexa of the uterus
amnion
areola

Vocabulary (Slide 2 of 34)

Term	Meaning/Definition
adnexa of the uterus	Fallopian tubes, ovaries, and supporting ligaments
amnion	Innermost membranous sac surrounding the developing fetus
areola	Dark-pigmented area surrounding the breast nipple

Vocabulary (Slide 3 of 34)

Term
Bartholin glands
cervix
chorion

Vocabulary (Slide 4 of 34)

Term	Meaning/Definition
Bartholin glands	Small, mucus-secreting exocrine glands at the vaginal orifice; named for Caspar Bartholin, a Danish anatomist who described them in 1637
cervix	Lower, neck-like portion of the uterus
chorion	Outermost layer of the two membranes surrounding the embryo

Vocabulary (Slide 5 of 34)

Term
clitoris
coitus
corpus luteum

Vocabulary (Slide 6 of 34)

Term	Meaning/Definition
clitoris	Sensitive erectile tissue anterior to the opening of the female urethra
coitus	Sexual intercourse; copulation
corpus luteum	Empty ovarian follicle; secretes progesterone after release of egg cell; means “yellow body”

Vocabulary (Slide 7 of 34)

Term
cul-de-sac
embryo
endometrium

Vocabulary (Slide 8 of 34)

Term	Meaning/Definition
cul-de-sac	Region in lower abdomen, midway between the rectum and the uterus
embryo	Stage in prenatal development from 2 to 8 weeks
endometrium	Inner, mucous membrane lining of the uterus

Vocabulary (Slide 9 of 34)

Term
estrogen
fallopian tube
fertilization

Vocabulary (Slide 10 of 34)

Term	Meaning/Definition
estrogen	Hormone produced by the ovaries that promotes female secondary sex characteristics
fallopian tube	One of a pair of ducts through which the ovum travels to the uterus; oviduct; named for Gabriello Falloppia, an Italian anatomist
fertilization	Union of sperm cell and ovum from which the embryo develops

Vocabulary (Slide 11 of 34)

Term
fetus
fimbriae (<i>sing.</i> : fimbria)
follicle-stimulating hormone (FSH)

Vocabulary (Slide 12 of 34)

Term	Meaning/Definition
fetus	Stage in prenatal development from 8 to 39 or 40 weeks
fimbriae (<i>sing.</i> : fimbria)	Finger- or fringe-like projections at the end of the fallopian tubes
follicle-stimulating hormone (FSH)	Secreted by the pituitary gland to stimulate maturation of the ovum

Vocabulary (Slide 13 of 34)

Term
gamete
genitalia
gestation

Vocabulary (Slide 14 of 34)

Term	Meaning/Definition
gamete	Male or female sexual reproductive cell; sperm cell or ovum
genitalia	Reproductive organs; also called genitals
gestation	Period from fertilization of the ovum to birth

Vocabulary (Slide 15 of 34)

Term
gonad
gynecology
human chorionic gonadotropin (hCG)

Vocabulary (Slide 16 of 34)

Term	Meaning/Definition
gonad	Female or male reproductive organ that produces sex cells and hormones; ovary or testis
gynecology	Study of the female reproductive organs, including the breasts
human chorionic gonadotropin (hCG)	Hormone produced by the placenta to sustain pregnancy by stimulating ovaries to produce estrogen and progesterone

Vocabulary (Slide 17 of 34)

Term
hymen
labia
lactiferous ducts

Vocabulary (Slide 18 of 34)

Term	Meaning/Definition
hymen	Mucous membrane partially or completely covering the opening to the vagina
labia	Lips of the vagina
lactiferous ducts	Tubes that carry milk within the breast

Vocabulary (Slide 19 of 34)

Term
luteinizing hormone (LH)
mammary papilla
menarche

Vocabulary (Slide 20 of 34)

Term	Meaning/Definition
luteinizing hormone (LH)	Secreted by the pituitary gland to promote ovulation
mammary papilla	Nipple of the breast
menarche	Beginning of the first menstrual period and ability to reproduce

Vocabulary (Slide 21 of 34)

Term
menopause
menstruation
myometrium
neonatology

Vocabulary (Slide 22 of 34)

Term	Meaning/Definition
menopause	Gradual ending of menstruation
menstruation	Monthly shedding of the uterine lining
myometrium	Muscle layer of the uterus
neonatology	Branch of medicine that studies the disorders and care of the newborn (neonate)

Vocabulary (Slide 23 of 34)

Term
obstetrics
orifice
ovarian follicle

Vocabulary (Slide 24 of 34)

Term	Meaning/Definition
obstetrics	The branch of medicine concerned with pregnancy and childbirth
orifice	An opening
ovarian follicle	Developing sac enclosing each ovum within the ovary

Vocabulary (Slide 25 of 34)

Term
ovary
ovulation
ovum (<i>pl.:</i> ova)

Vocabulary (Slide 26 of 34)

Term	Meaning/Definition
ovary	One of a pair of female organs on each side of the pelvis; almond-shaped; produced egg cells and hormones
ovulation	Release of the ovum from the ovary
ovum (<i>pl.</i> : ova)	Mature egg cell (female gamete)

Vocabulary (Slide 27 of 34)

Term
parturition
perineum
pituitary gland

Vocabulary (Slide 28 of 34)

Term	Meaning/Definition
parturition	Act of giving birth
perineum	Area between the anus and the vagina in females
pituitary gland	Endocrine gland at the base of the brain that produces hormones to stimulate the ovaries; also regulates other endocrine organs

Vocabulary (Slide 29 of 34)

Term
placenta
pregnancy
progesterone

Vocabulary (Slide 30 of 34)

Term	Meaning/Definition
placenta	Vascular organ attached to uterine wall during pregnancy; allows exchange of oxygen, nutrients, and fetal waste products between mother and fetus
pregnancy	Condition of having a developing embryo and fetus in female uterus for about 40 weeks
progesterone	Hormone produced by corpus luteum in ovary and placenta of pregnant women

Vocabulary (Slide 31 of 34)

Term
puberty
uterine serosa
uterus

Vocabulary (Slide 32 of 34)

Term	Meaning/Definition
puberty	Point in life cycle at which sex characteristics appear and gametes are produced
Perimetrium (uterine serosa)	Outermost layer surrounding the uterus
uterus	Hollow, pear-shaped muscular female organ in which embryo and fetus develop, and from which menstruation occurs

Vocabulary (Slide 33 of 34)

Term
vagina
vulva
zygote

Vocabulary (Slide 34 of 34)

Term	Meaning/Definition
vagina	Muscular, mucosa-lined canal extending from the uterus to the exterior of the body
vulva	External female genitalia; includes the labia, hymen, clitoris, and vaginal orifice
zygote	The stage in prenatal development from fertilization and implantation to 2 weeks

QUICK QUIZ (Slide 4 of 5)

4. Which term refers to the inner, mucous membrane lining of the uterus?

- A. Endometrium
- B. Fallopian tube
- C. Chorion
- D. Amnion

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 1 of 6)

Combining Form	Meaning
amni/o	amnion
bartholin/o	Bartholin gland
cervic/o	cervix, neck
chori/o	chorion
chorion/o	chorion
colp/o	vagina

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 2 of 6)

Combining Form	Meaning
culd/o	cul-de-sac
episi/o	vulva
galact/o	milk
gynec/o	woman, female
hyster/o	uterus, womb

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 3 of 6)

Combining Form	Meaning
lact/o	milk
mamm/o	breast
mast/o	breast
men/o	menses, menstruation
metr/o	uterus

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 4 of 6)

Combining Form	Meaning
my/o	muscle
myom/o	muscle tumor
nat/i	birth
obstetr/o	midwife
o/o	egg

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 5 of 6)

Combining Form	Meaning
oophor/o	ovary
ov/o	egg
ovari/o	ovary
ovul/o	egg
perine/o	perineum

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 6 of 6)

Combining Form	Meaning
phor/o	to bear
salping/o	fallopian tubes
uter/o	uterus
vagin/o	vagina
vulv/o	vulva

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 1 of 2)

Suffix	Meaning
-arche	beginning
-cyesis	pregnancy
-gravida	pregnancy
-parous	to bear, bring forth

Terminology: Combining Forms, Suffixes, and Prefixes (Slide 2 of 2)

Suffix	Meaning
-rrhea	discharge
-salpinx	uterine tube
-tocia	labor, birth
-version	act of turning

Terminology: Combining Forms, Suffixes, Prefixes (Slide 1 of 2)

Prefix	Meaning
dys-	painful
endo-	within
in-	in
intra-	within
multi-	many

Terminology: Combining Forms, Suffixes, & Prefixes (Slide 2 of 2)

Prefix	Meaning
nulli-	no, not, none
pre-	before
primi-	first
retro-	backward

QUICK QUIZ (Slide 5 of 5)

5. Which term means removal of the breast?

- A. Hysterectomy
- B. Mastectomy
- C. Mastitis
- D. Mammoplasty

Chapter 8

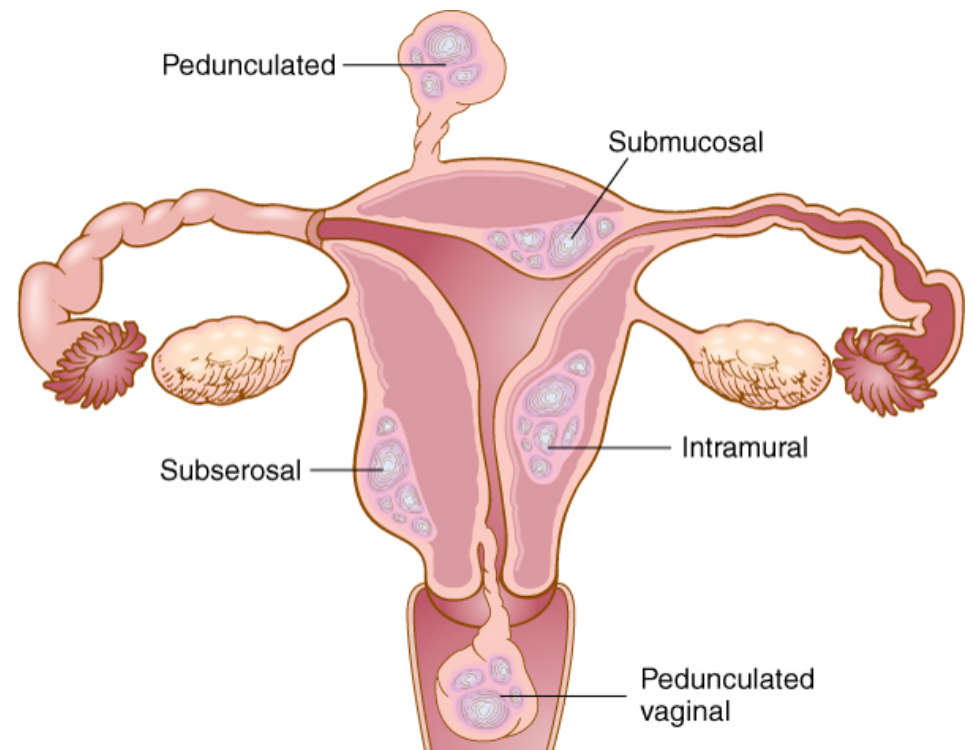
Lesson 8.2

Pathology: Uterus (slide 1 of 2)

- Cervical cancer
 - Malignant cells within the cervix
- Cervicitis
 - Inflammation of the cervix
- Endometrial cancer
 - Malignant tumor of the uterine lining
- Endometriosis
 - Endometrial tissue located outside the uterus

Pathology: Uterus (slide 2 of 2)

- Fibroids are benign tumors of the uterus



Pathology: Gynecological

Ovaries

- Ovarian cancer
 - Malignant tumor of the ovary
- Ovarian cysts
 - Collections of fluid within a sac in the ovary

Fallopian tubes

- Pelvic inflammatory disease
 - Examples: salpingitis, oophoritis, endometritis
 - Leading causes of PID are sexually transmitted diseases

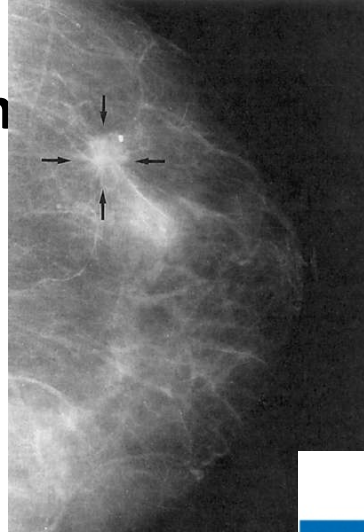
Pathology: Breast (slide 1 of 2)

- Breast
 - Breast cancer
 - Fibrocystic disease

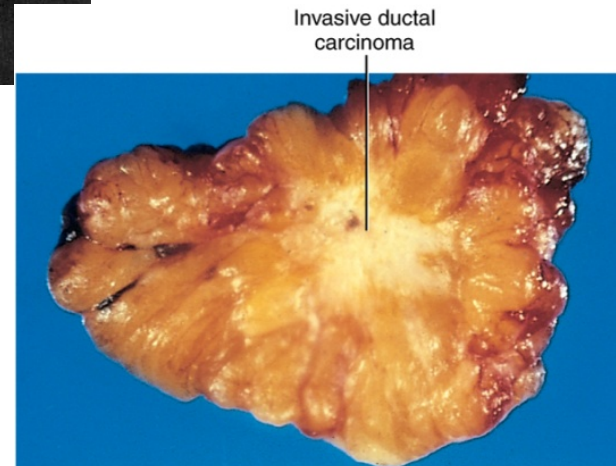
Pathology: Breast (slide 2 of 2)

Breast Cancer

Arrows in mammogram point to invasive carcinoma of the breast.



Cut section of invasive ductal carcinoma of the breast.



Chapter 8

Lesson 8.3

Pathology: Pregnancy

- Abruptio placentae
 - Premature separation of the normally implanted placenta
- Choriocarcinoma
 - Malignant tumor of the placenta
- Ectopic pregnancy
 - Misplaced implantation of egg
- Placenta previa
- Preeclampsia

Pathology: Neonatal (slide 1 of 2)

- Down syndrome
- Hemolytic disease of the newborn
- Hyaline membrane disease
- Hydrocephalus
- Meconium aspiration syndrome
- Pyloric stenosis

Pathology: Neonatal (slide 2 of 2)

- **APGAR scoring chart**
 - Assessment of newborn
 - Scoring 1 to 5 min after birth

APGAR SCORING CHART

SIGN	0	1	2
Heart rate	Absent	Below 100	Over 100
Respiratory effort	Absent	Slow, irregular	Good, crying
Muscle tone	Limp	Some flexion of extremities	Active motion
Response to catheter in nostril (tested after oropharynx is clear)	No response	Grimace	Cough or sneeze
Color	Blue, pale	Body pink, extremities blue	Completely pink

Clinical Tests and Procedures

Clinical tests

- Pap test (Pap smear)
- Pregnancy test

Procedures

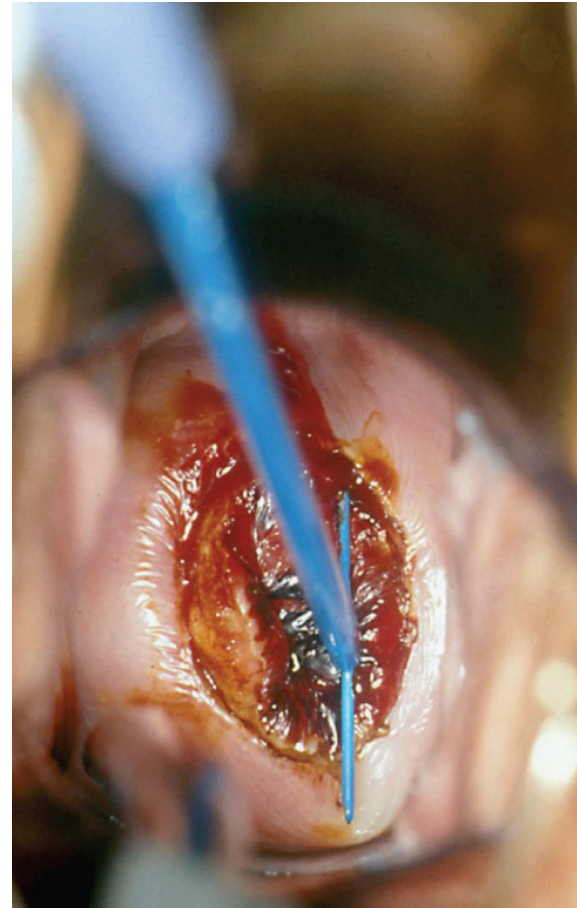
- Hystersalpingography
- Mammography
- Breast ultrasound/breast MRI
- Pelvic ultrasonography

Clinical Procedures (Slide 1 of 2)

Gynecological procedures

- **Conization**

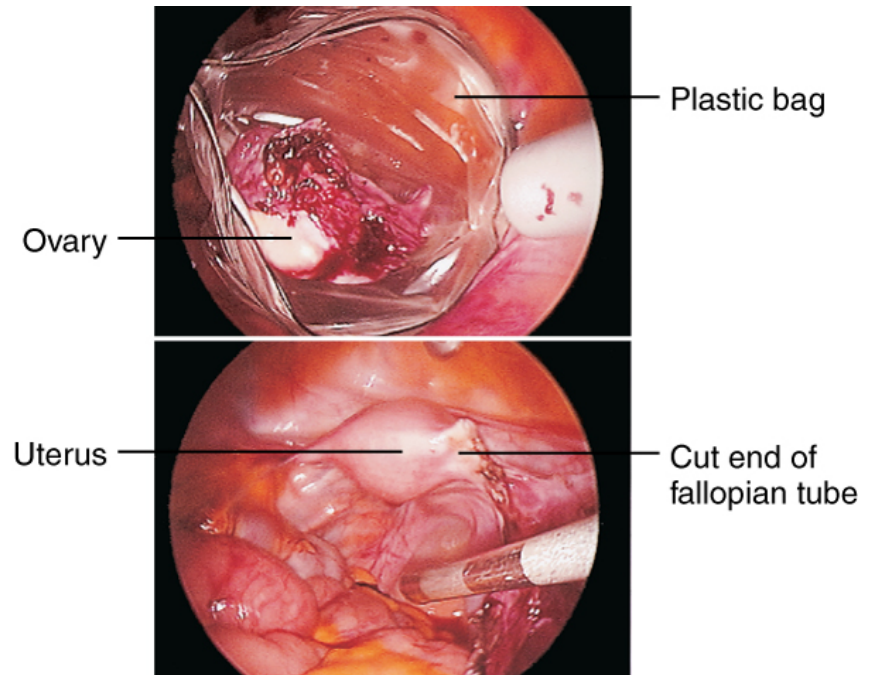
- Removal of a cone-shaped section of the cervix.
- Figure shows LEEP for cone biopsy



Clinical Procedures (Slide 2 of 2)

Gynecological procedures

- **Laparoscopic oophorectomy**
 - A, Ovary placed in plastic bag
 - B, Extracted through the laparoscope



Clinical Tests & Procedures

Procedures

- **Procedures during pregnancy**
 - **Amniocentesis**
Needle placement guided by ultrasound images



Abbreviations (Slide 1 of 8)

Abbreviation

AB

AFP

BRCA1/BRCA2

BSE

CA 125

C-section, CS

CIN

CIS

CVS

Cx

D & C

Abbreviations (Slide 2 of 8)

Abbreviation	Meaning
AB	Abortion
AFP	Alpha-fetoprotein-high levels in amniotic fluid of fetus or maternal serum
BRCA1/BRCA2	Breast cancer 1 and 2 - genetic mutations associated with increased risk for breast cancer
BSE	Breast self-examination
CA 125	Protein marker elevated in ovarian cancer
C-section, CS	Cesarean section
CIN	Cervical intraepithelial neoplasia
CIS	Carcinoma in situ
CVS	Chorionic villus sampling
Cx	Cervix
D & C	Dilation (dilation and curettage)

Abbreviations (Slide 3 of 8)

Abbreviation

DCIS

DUB

FHR

FSH

G

GnRH

GYN

hCG OR HCG

HDN

HPV

HRT

HSG

IUD

Abbreviations (Slide 4 of 8)

Abbreviation	Meaning
DCIS	Ductal carcinoma in situ
DUB	Dysfunctional uterine bleeding
FHR	Fetal heart rate
FSH	Follicle-stimulating hormone
G	Gravida (pregnant)
GnRH	Gonadotropin-releasing hormone
GYN	Gynecology
hCG OR HCG	Human chorionic gonadotropin
HDN	Hemolytic disease of the newborn
HPV	Human papillomavirus
HRT	Hormone replacement therapy
HSG	Hysterosalpingography
IUD	Intrauterine device; contraceptive

Abbreviations (Slide 5 of 8)

Abbreviation

IVF

LEEP

LH

LMP

multip

OB

para 2-0-1-2

Pap test

PID

PMS

Abbreviations (Slide 6 of 8)

Abbreviation	Meaning
IVF	In vitro fertilization
LEEP	Loop electrocautery excision procedure
LH	Luteinizing hormone
LMP	Last menstrual period
multip	Multipara; multiparous
OB	Obstetrics
para 2-0-1-2	A woman's reproductive history: 2 full-term infants, 0 preterm, 1 abortion, and 2 living children
Pap test	Papanicolaou smear-test
PID	Pelvic inflammatory disease
PMS	Premenstrual syndrome

Abbreviations (Slide 7 of 8)

Abbreviation

primip

SLN biopsy or SNB

TAH-BSO

UAE

VH

Abbreviations (Slide 8 of 8)

Abbreviation	Meaning
primip	Primipara; primiparous
SLN biopsy or SNB	Sentinel lymph node biopsy
TAH-BSO	Total abdominal hysterectomy with bilateral salpingo-oophorectomy
UAE	Uterine artery embolization
VH	Vaginal hysterectomy

Review Sheet – Combining Forms

(Slide 1 of 12)

Combining Form
amni/o
bartholin/o
cephal/o
cervic/o
chori/o, chorion/o
colp/o

Review Sheet – Combining Forms

(Slide 2 of 12)

Combining Form	Meaning
amni/o	amnion
bartholin/o	Bartholin gland
cephal/o	head
cervic/o	cervix, neck
chori/o, chorion/o	chorion
colp/o	vagina

Review Sheet – Combining Forms

(Slide 3 of 12)

Combining Form
culd/o
episi/o
galact/o
gynec/o
hyster/o

Review Sheet – Combining Forms

(Slide 4 of 12)

Combining Form	Meaning
culd/o	cul-de-sac
episi/o	vulva
galact/o	milk
gynec/o	woman, female
hyster/o	uterus, womb

Review Sheet – Combining Forms

(Slide 5 of 12)

Combining Form
lact/o
mamm/o
mast/o
men/o
metr/o

Review Sheet – Combining Forms

(Slide 6 of 12)

Combining Form	Meaning
lact/o	milk
mamm/o	breast
mast/o	breast
men/o	menses, menstruation
metr/o	uterus

Review Sheet – Combining Forms

(Slide 7 of 12)

Combining Form
my/o
myom/o
nat/i
obstetr/o
olig/o
o/o

Review Sheet – Combining Forms

(Slide 8 of 12)

Combining Form	Meaning
my/o	muscle
myom/o	muscle tumor
nat/i	birth
obstetr/o	pregnancy and childbirth
olig/o	scanty
o/o	egg

Review Sheet – Combining Forms

(Slide 9 of 12)

Combining Form
oophor/o
ov/o
ovari/o
ovul/o
perine/o

Review Sheet – Combining Forms

(Slide 10 of 12)

Combining Form	Meaning
oophor/o	ovary
ov/o	egg
ovari/o	ovary
ovul/o	egg
perine/o	perineum

Review Sheet – Combining Forms

(Slide 11 of 12)

Combining Form
phor/o
py/o
salping/o
uter/o
vagin/o
vulv/o

Review Sheet – Combining Forms

(Slide 12 of 12)

Combining Form	Meaning
phor/o	to bear
py/o	pus
salping/o	fallopian tubes
uter/o	uterus
vagin/o	vagina
vulv/o	vulva

Review Sheet – Suffixes

(Slide 1 of 10)

Suffix
-arche
-cele
-cyesis
-ectasis
-ectomy

Review Sheet – Suffixes

(Slide 2 of 10)

Suffix	Meaning
-arche	beginning
-cele	hernia
-cyesis	pregnancy
-ectasis	stretching, dilation
-ectomy	removal, excision

Review Sheet – Suffixes

(Slide 3 of 10)

Suffix
-flexion
-genesis
-gravida
-itis
-pareunia
-parous

Review Sheet – Suffixes

(Slide 4 of 10)

Suffix	Meaning
-flexion	flexible
-genesis	producing; forming
-gravida	pregnancy
-itis	inflammation
-pareunia	sexual intercourse
-parous	to bear, bring forth

Review Sheet – Suffixes

(Slide 5 of 10)

Suffix
-plasia
-plasty
-ptosis
-rrhagia
-rrhaphy

Review Sheet – Suffixes

(Slide 6 of 10)

Suffix	Meaning
-plasia	development, formation
-plasty	surgical repair
-ptosis	dropping, sagging
-rrhagia	bursting forth (blood)
-rrhaphy	suture

Review Sheet – Suffixes

(Slide 7 of 10)

Suffix
-rrhea
-salpinx
-scopy
-stenosis
-stomy

Review Sheet – Suffixes

(Slide 8 of 10)

Suffix	Meaning
-rrhea	discharge
-salpinx	uterine tube
-scopy	visual examination
-stenosis	tightening
-stomy	new opening

Review Sheet – Suffixes

(Slide 9 of 10)

Suffix
-tocia
-tomy
-tresia
-version

Review Sheet – Suffixes

(Slide 10 of 10)

Suffix	Meaning
-tocia	labor, birth
-tomy	process of cutting
-tresia	opening
-version	act of turning

Review Sheet – Prefixes

(Slide 1 of 6)

Prefix
bi-
dys-
endo-
in-
intra-
multi-

Review Sheet – Prefixes

(Slide 2 of 6)

Prefix	Meaning
bi-	two
dys-	painful
endo-	within
in-	in
intra-	within
multi-	many

Review Sheet – Prefixes

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Prefix
nulli-
oxy-
peri-
pre-

Review Sheet – Prefixes

(Slide 4 of 6)

Prefix	Meaning
nulli-	no, not, none
oxy-	swift, sharp
peri-	surrounding
pre-	before

Review Sheet – Prefixes

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Prefix
primi-
pseudo-
retro-
uni-

Review Sheet – Prefixes

(Slide 6 of 6)

Prefix	Meaning
primi-	first
pseudo-	false
retro-	backward
uni-	one